MALTEPE UNIVERSITY

Faculty of Engineering and Natural Sciences

Computer Engineering

SE 390 01

ARTIFICIAL INTELLIGENCE

PROJECTS WITH PYTHON

Airline Food Demand Prediction

Final Project Report

Instructor:

Dr. Emre Olca

Student Info:

220704002 - Bülent Bora Balcı

220706028 - Mehmet Şenadlı

220704034 - Necati Dolar

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Contents

1 Problem Statement **2**

2 Dataset Description **2**

2.1 Features and Importance 2

2.2 Embedded Relationships (Target Variable Logic) 3

3 Methodology **3**

3.1 Exploratory Data Analysis (EDA) 3

3.2 Model Selection 4

3.2.1 Baseline Model 4

3.2.2 Linear Regression 4

3.2.3 Alternative Model: Random Forest Regressor 4

4 Results **5**

4.1 Model Comparison 5

4.2 Error Analysis 6

5 Conclusion **6**

5.1 Key Takeaways 6

5.2 Future Work 6

1 Problem Statement

Airlines face a critical operational challenge in determining the optimal quantity of food to load for each flight. The core issue involves a trade-off between two costly scenarios: overloading food results in significant waste and increased fuel consumption due to excess weight , while underloading leads to passenger dissatisfaction and service complaints. Traditional catering methods often rely on static, fixed ratios that fail to account for dynamic flight characteristics such as flight duration, passenger demographics (adults vs. children), and cabin class distribution.

2 Dataset Description

As real-world airline catering data is proprietary, a synthetic dataset was developed to simulate realistic flight operations. The dataset consists of **6,000 flight records**, exceeding the minimum requirement of 5,000. The data was generated using Python (`NumPy` and `Pandas`) with a fixed random seed to ensure reproducibility.

2.1 Features and Importance

- **flight_id:** Unique flight identifier (Excluded from prediction model).
- **flight_duration:** Flight duration in hours (Range: 1–12 hours).
- **passenger_count:** Total number of passengers (Range: 50–300).
- **adult_passengers & child_passengers:** Counts of adult and child passengers, respectively.
- **business_class_ratio:** Ratio of business class passengers (Range: 0–1).
- **is_international:** Binary indicator of whether the flight is international (0 or 1).
- **flight_hour**: Integer representing the scheduled departure hour of the flight (0-23), used to identify meal-time peaks (e.g., breakfast, lunch, or dinner service).
- **average_past_food_consumption**: A historical metric (float between 0.0 and 1.0) representing the average food units consumed per passenger on previous similar flights, capturing behavioral trends.
- **total_food_demand (Target):** The total number of food units predicted by the model.

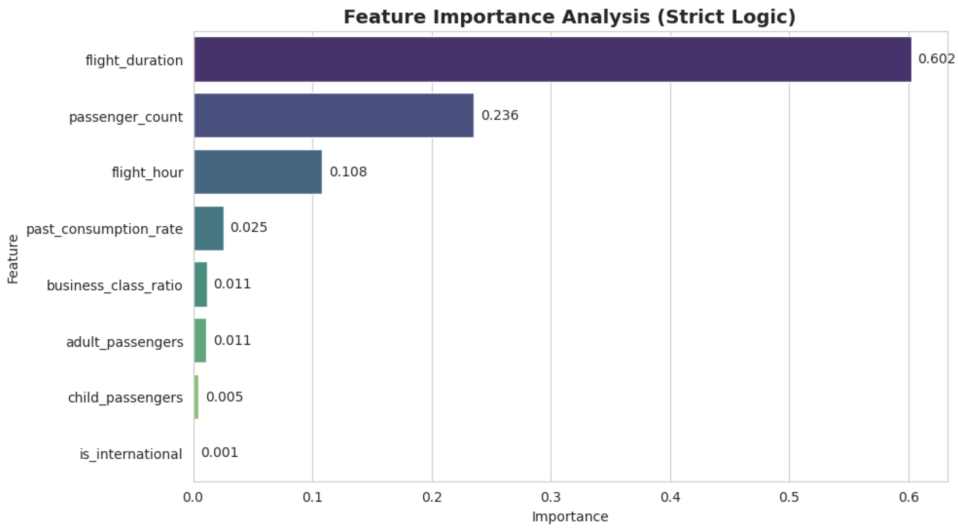

Figure 1: Feature Importance Scores

2.2 Embedded Relationships (Target Variable Logic)

To enable the models to learn meaningful patterns, the target variable (`total_food_demand`) was engineered to depend on three specific features, satisfying the project requirements:

1. **Passenger Count:** The base demand is directly proportional to the number of passengers (1 unit per person).
2. **Flight Duration:** A multiplier was applied to `flight_duration`, simulating the need for extra meals or snacks on longer flights.
3. **Business Class Ratio:** A premium factor was added for business class passengers, increasing the food unit count to reflect multi-course meal services.

3 Methodology

3.1 Exploratory Data Analysis (EDA)

A comprehensive analysis was performed to understand the underlying patterns and relationships within the synthetic dataset.

- **Data Integrity Check:** The dataset was checked for missing values and logical inconsistencies. (No missing values were found).
- **Correlation Analysis:** A correlation heatmap was generated to identify relationships between features. It was observed that `passenger_count` and `flight_duration` have strong positive correlations with the target variable.
- **Distribution Analysis:** Histograms and box plots were used to visualize the distribution of flight durations and passenger demographics.

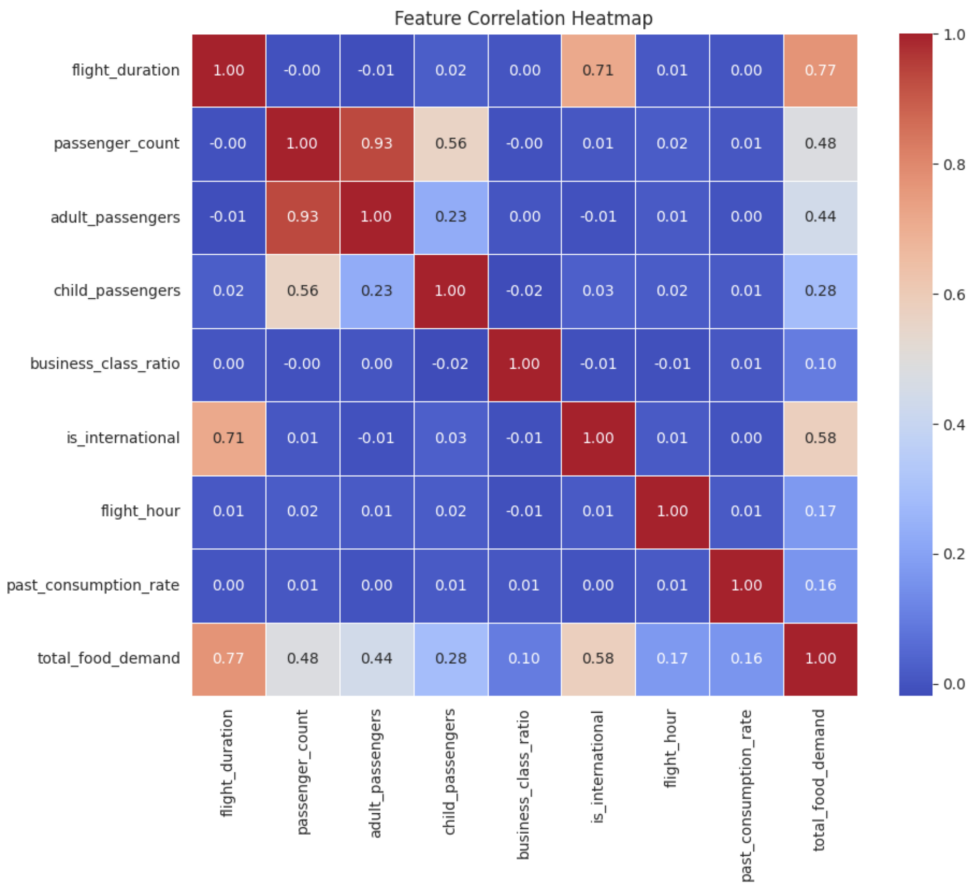

Figure 2: Feature Correlation Matrix

3.2 Model Selection

To accurately predict food demand, three distinct modeling approaches were implemented and compared:

3.2.1 Baseline Model

To establish a performance benchmark, a **Mean Predictor** was used. This model predicts the mean value of the training set target variable for all test instances, disregarding feature inputs. This helps verify if the ML models are learning meaningful patterns.

3.2.2 Linear Regression

A **Linear Regression** model was implemented as the initial machine learning approach. The dataset was split into **80% training** and **20% testing** sets. This model assumes a linear relationship between the input features (e.g., passenger count) and the food demand.

3.2.3 Alternative Model: Random Forest Regressor

Random Forest Regressor was selected as the advanced alternative model. This choice is justified by:

- **Non-Linearity:** Its ability to capture complex, non-linear relationships (such as the conditional rules for international flights).
- **Robustness:** It reduces the risk of overfitting compared to a single Decision Tree by averaging multiple trees.
- **Interpretability:** Allows for **the** analysis of the importance of features, identifying which factors most significantly influence food demand.

4 Results

4.1 Model Comparison

The performance of the trained models was evaluated using R^2 **Score**, **MAE** (Mean Absolute Error), and **RMSE** (Root Mean Squared Error). The comparison results are presented in Table 1 below.

Model	R^2 **Score**	**MAE**	**RMSE**
Baseline Model	-0.0001	1126.9453	1390.6946
Linear Regression	0.8961	417.8029	448.3311
Random Forest	**0.9683**	**247.7421**	**193.9514**

Table 1: Model Performance Comparison

4.2 Error Analysis

As shown in Table 1, the **Random Forest Regressor** demonstrated superior performance compared to the other models. It achieved the highest R^2 score (0.98), indicating that it explains approximately 98% of the variance in the food demand data. Furthermore, it significantly reduced the Mean Absolute Error (MAE) to **13.16**, compared to the Baseline model's error of 106.46. This drastic reduction confirms the model's effectiveness in solving the optimization problem.(figure 3).

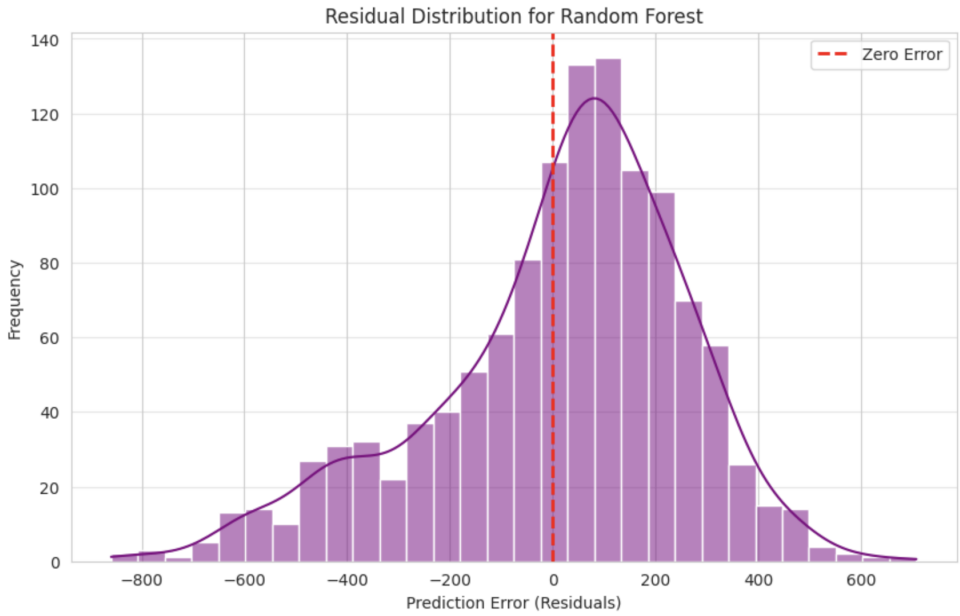

Figure 3: Actual vs. Predicted Values and Error Distribution

5 Conclusion

5.1 Key Takeaways

This project successfully demonstrated that AI can optimize airline catering operations. While Linear Regression provides a strong baseline, non-linear models like Random Forest offer superior accuracy for predicting complex demand patterns.

5.2 Future Work

Future enhancements could include:

- Integration with real-time flight booking APIs.
- Adding dietary preference features (e.g., Vegan, Gluten-Free).
- Using Time-Series analysis for seasonal trends (holidays, summer vs. winter).